

Relapse and Management After Treatment

One group reported relapses beginning 6 years after completion of 2 years of triple drug therapy.[54] Two groups reported relapse after smear negativity had occurred.[54,55] A 1-year course of treatment was associated with a higher relapse rate compared to a 2-year course.[53] These relapses are not considered to be re-infections. There is variation in opinion on how best to manage this issue. The WHO strategy relies on patient self-referral for follow-up. In contrast, USPHS recommendations permit resumption of treatment based on physician evaluation. Another approach is to continue dapsone monotherapy indefinitely after completing multidrug therapy.

Alternative Antibiotics and Potential Regimens

In addition to rifampin, several antibiotics have potent bactericidal activity against *Mycobacterium leprae*, including minocycline, clarithromycin, and the fluoroquinolones (ofloxacin and levofloxacin). A single-dose regimen of rifampin, ofloxacin, and minocycline is being studied for paucibacillary disease, though the long-term relapse rate remains undetermined.[56] However, daily use of these agents is prohibitively expensive in regions where leprosy is endemic and healthcare resources are limited. It is also unclear whether combinations of these potent bactericidal drugs—though theoretically appealing by analogy to tuberculosis treatment—offer any advantage over current WHO or USPHS guidelines, although they were well tolerated in a small trial.[57]

Practical Treatment Approach in Multibacillary Leprosy

Given the low incidence of primary dapsone resistance in our multibacillary patient population,[58] we often use a combination of rifampin (600 mg daily) and dapsone (100 mg daily) for 2 to 3 years. After completion of this regimen, the decision to continue dapsone indefinitely or accept an uncertain relapse risk is discussed with the patient.

Management of Reversal Reactions

In reversal reactions (type 1 reactions), prompt initiation of prednisone therapy is recommended due to the risk of permanent nerve damage. The typical starting dose is 0.5 to 1.0 mg/kg/day. The dose should be titrated based on nerve tenderness, the patient’s symptoms, and careful sensory assessment of hands and feet—such as with graded Semmes-Weinstein filaments.

Prednisone therapy should be tapered slowly, and patients should be informed that treatment may last 6 months or longer. If neuritis does not respond promptly to prednisone, rest and splinting of affected extremities are advised.

Box 186-2: Medical Management of Reaction States

Reaction Type	Thalidomide	Prednisone or Prednisolone	Duration	Other Agents of Unproven Value
Reversal reactions (type 1)	Of no value	0.5–1.0 mg/kg/day. Rifampin may increase catabolism. Taper slowly. Alternate-day therapy may be well tolerated.	Usually 6 months to 2 years; may vary	Nonsteroidal anti-inflammatory agents
Erythema nodosum leprosum (type II)	Most effective if available and not contraindicated. Initial dose: 100–200 mg once daily at bedtime. Maintenance: 50 mg every other day to 500 mg daily	Used if thalidomide is unavailable: 0.5–1.0 mg/kg/day	Median duration ~5 years; may last up to 10 years	Pentoxifylline
Lucio phenomenon	Of no value	May be helpful	—	Plasmapheresis reported beneficial in refractory cases